

METIMMOX-1 Value of Information Analysis

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

Ben Geisler

2025-10-16

Table of contents

1	Summary	1
2	Results	2
2.1	Strategy Performance Comparison	2
2.2	Expected Value of Perfect Information	3
3	Interpretation	4

1 Summary

This report presents an Expected Values of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 economic model. I aim to use EVPPIs as probabilistic sensitivity analyses to identify parameters that contribute most to decision uncertainty and prioritize future research efforts. The report includes:

- Expected Value of Perfect Information (EVPI) calculation and decision uncertainty assessment
- Strategy performance comparison across treatment alternatives
- Incremental cost-effectiveness scatter plot analysis
- Interpretation of findings for clinical and research decision-making

2 Results

2.1 Strategy Performance Comparison

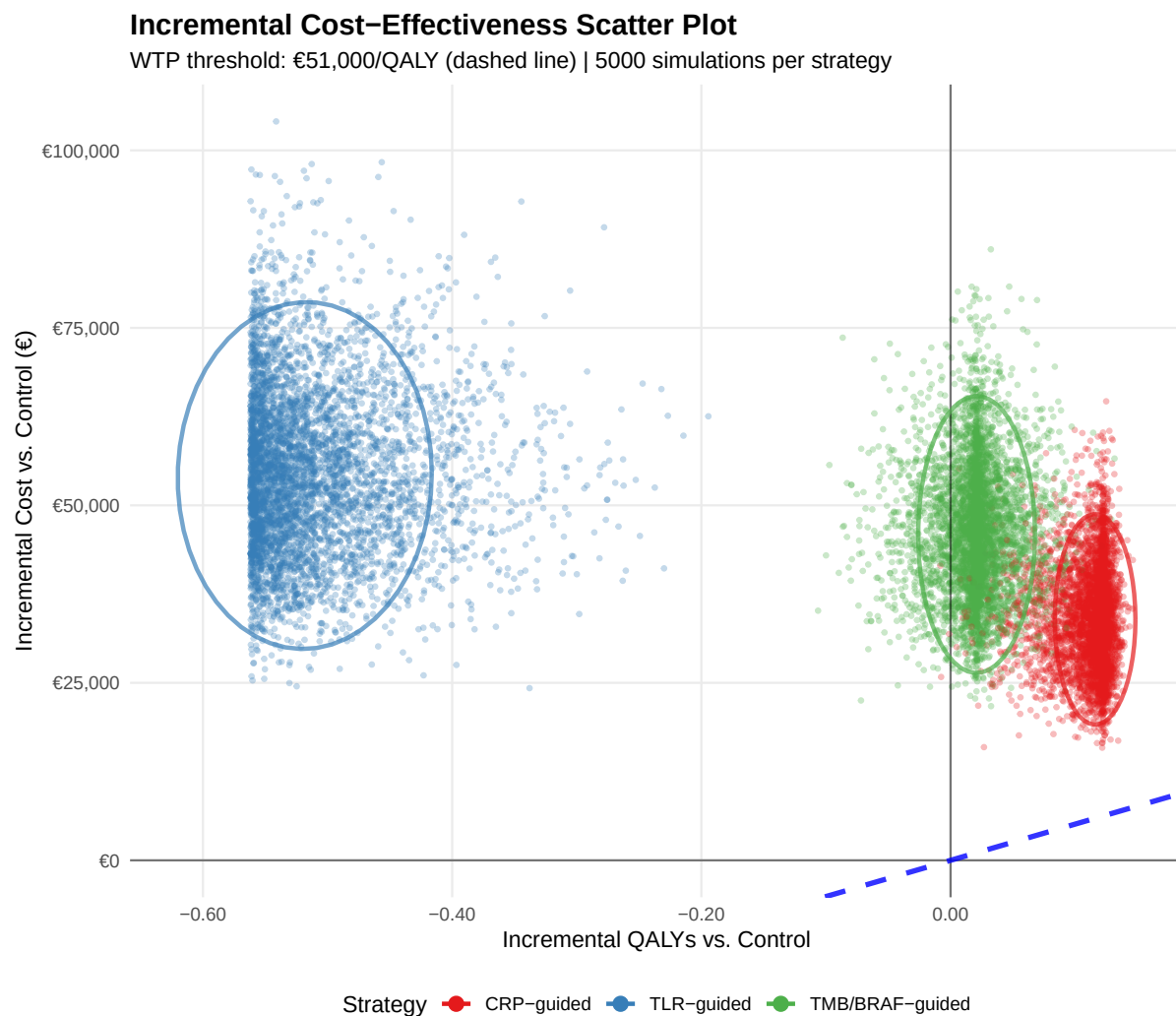


Figure 1: Incremental Cost-Effectiveness Scatter Plot

Table 1: Strategy Performance Summary: Mean (Standard Error)

	Strategy	Absolute Values		Incremental vs. Control			ICER (€/QALY)
		Cost (€)	QALYs	NMB (€)	Inc. Cost (€)	Inc. QALYs	
control	Control	21740 (40)	1.805 (0.003)	70293 (144)	0	0.000	Dominated
crp	CRP-guided	56005 (107)	1.916 (0.003)	41730 (182)	34266	0.112	€ 306,423
tlr	TLR-guided	76470 (168)	1.296 (0.002)	-10390 (196)	54730	-0.509	Dominated
tmb_braf	TMB/BRAF-guided	68047 (136)	1.825 (0.003)	25022 (196)	46307	0.020	€2,277,732

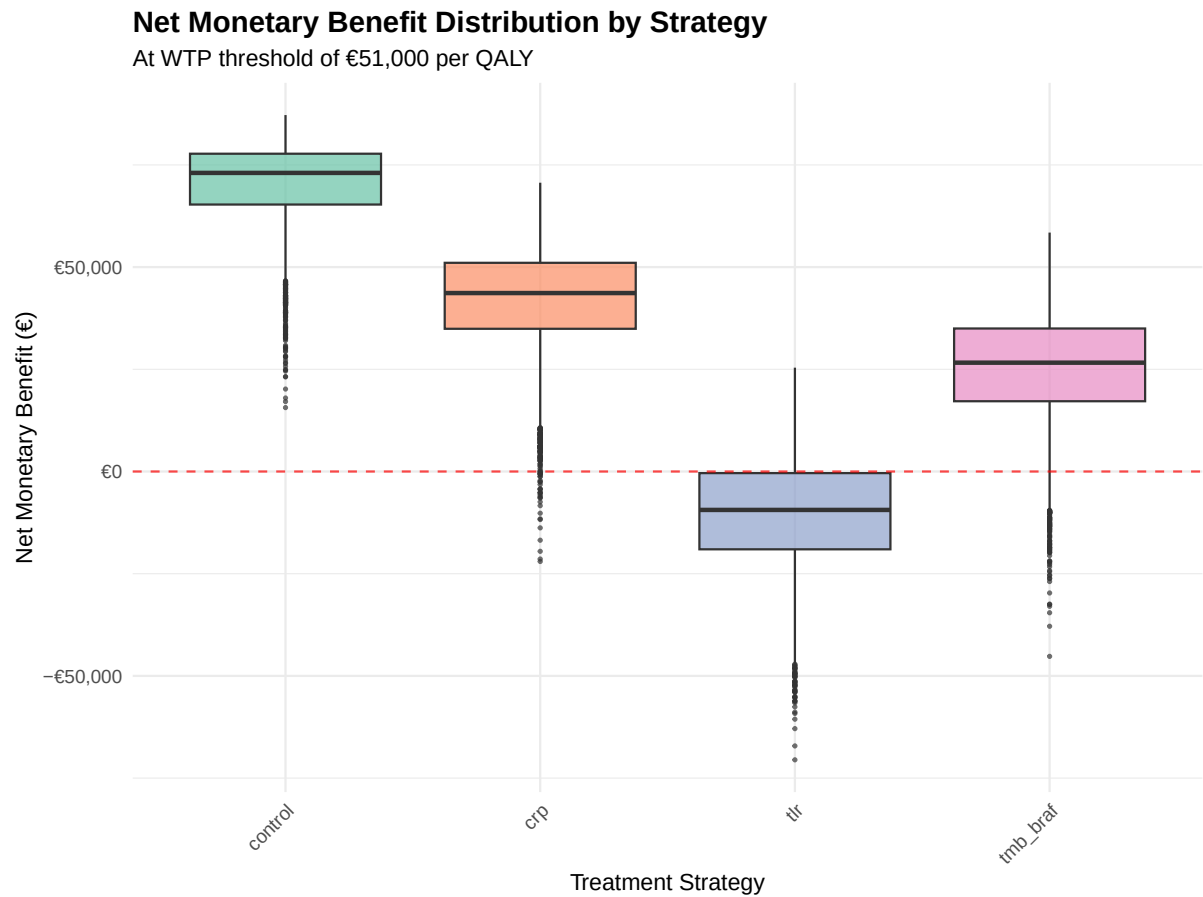


Figure 2: Net Monetary Benefit Distribution by Strategy

2.2 Expected Value of Perfect Information

Total EVPI: € 0

Expected value with perfect information: € 0

Expected value with current information: € 70292.51

Optimal strategy: control (standard care)

Decision certainty: Control strategy optimal in 100 % of simulations

The EVPI of € 0 indicates that there is no decision uncertainty in the current model. The standard care strategy (FLOX chemotherapy alone) dominates all biomarker-guided strategies across the entire range of parameter uncertainty considered in the probabilistic sensitivity analysis.

3 Interpretation

The analysis demonstrates that standard FLOX chemotherapy alone provides superior cost-effectiveness compared to biomarker-guided immunotherapy strategies at the €51,000/QALY threshold. The control strategy dominates across all parameter uncertainty ranges considered, indicating that biomarker strategies add substantial costs without sufficient QALY gains to justify the expense.

The biomarker-guided strategies show incremental costs ranging from € 34,266 to € 54,730 compared to standard care, while providing mixed effectiveness outcomes. Notably, the TLR-guided strategy shows negative expected net monetary benefit, indicating it is not cost-effective even compared to no intervention.

Given the absence of decision uncertainty ($EVPI = €0$), traditional EVPPI analysis cannot identify parameters for prioritized research, as no additional parameter precision would change the optimal treatment decision. This finding suggests that research efforts should focus on implementation of standard care rather than parameter refinement studies. The current evidence base appears sufficient for robust decision-making regarding these specific biomarker strategies.

Future research might better target fundamental redesign of biomarker approaches, investigation of alternative cost structures, or development of entirely different treatment strategies that could achieve cost-effectiveness thresholds. The robust dominance of standard care across parameter uncertainty ranges indicates that substantial changes to model assumptions or intervention design would be required to alter the cost-effectiveness conclusion.

The zero EVPI finding depends on current model assumptions and parameter uncertainty ranges. Results may differ under alternative model structures, wider parameter distributions, different willingness-to-pay thresholds, or longer time horizons that capture extended immunotherapy benefits. The healthcare system perspective may not reflect broader societal value, and real-world implementation factors are not captured in the current analysis.

Report completed on: 2025-10-16

Repository: ben-geisler/METIMMOX-1

Report version: 1.0